AI INTERNSHIP REPORT

ON

ARTIFICIAL INTELLIGENCE PROJECTS

(Chatbot with Rule-Based Responses, Tic-Tac-Toe AI, Recommendation System)

Submitted To

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By

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CANDIDATE'S DECLARATION

I, Shivam Kushwaha, Roll No. 25SCS1003001526, do hereby declare that the internship report titled “AI Internship at CodSoft” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

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ACKNOWLEDGEMENT

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I would also like to thank the faculty of the School of Computer Science and Engineering, IILM University, for their continuous support and encouragement throughout this internship.

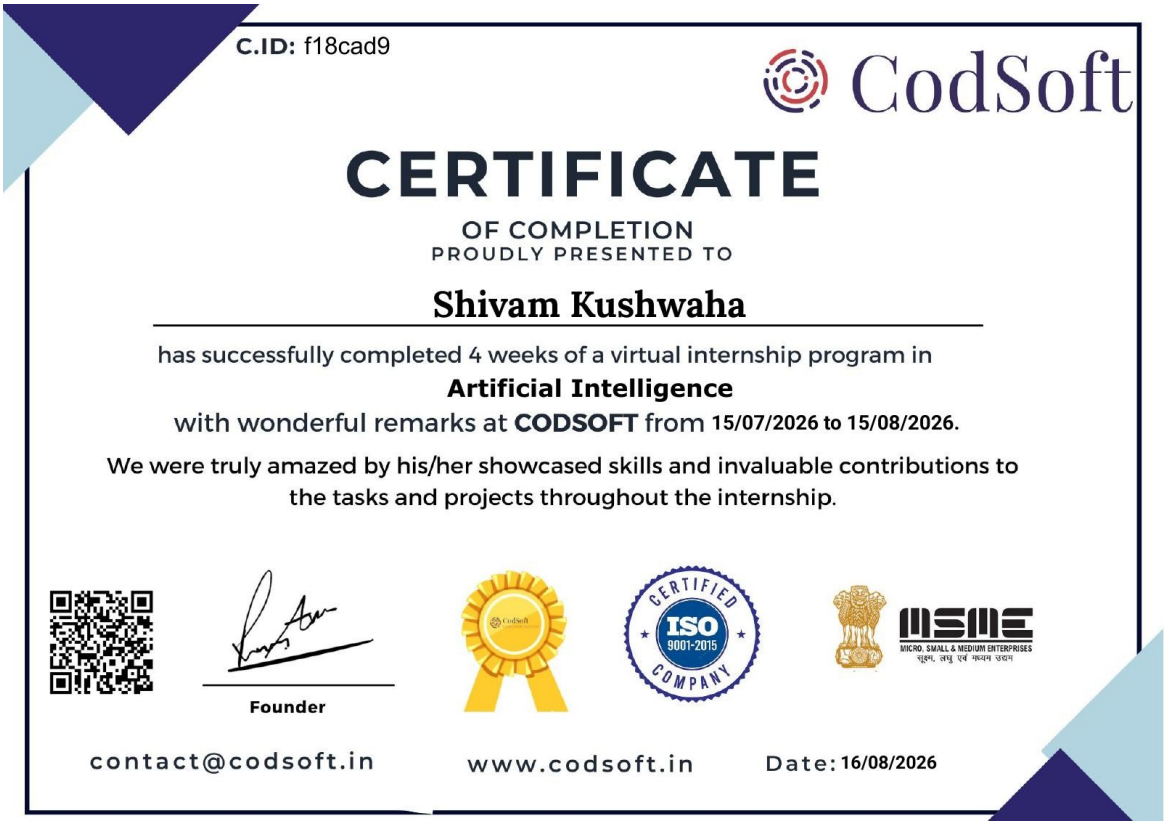
I express my gratitude to my parents for their blessings.

Signature

Student Name: Shivam Kushwaha

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INTERNSHIP COMPLETION CERTIFICATE



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CHAPTER 4: PROJECT DESCRIPTION

4.1 Introduction

Artificial Intelligence (AI) is one of the most rapidly growing fields of Computer Science, enabling machines to simulate human-like decision-making, learning, and reasoning. As part of my internship at CodSoft, I worked on a series of AI-based mini-projects designed to build a strong foundation in core AI concepts such as rule-based systems, search and game-playing algorithms, and recommendation techniques. This report presents the work carried out during the internship, including the problem statements, objectives, technologies used, methodology, and outcomes of each task.

4.2 Organization Profile

CodSoft is a technology-focused organization that provides internship and training programs to students in domains such as Artificial Intelligence, Web Development, Android Development, Python Programming, Java Programming and Data Science. It offers hands-on, project-based learning where interns are given real-world tasks to build practical skills. Through structured task sheets and mentorship, CodSoft helps students gain exposure to industry-relevant tools and problem-solving approaches, which was the primary mode of learning followed during this internship.

4.3 Problem Statement

Beginners in Artificial Intelligence often struggle to connect theoretical concepts, such as rule-based logic, search algorithms and recommendation techniques, with practical implementation. The internship addressed this gap by assigning three independent AI tasks, each targeting a specific concept: conversational logic (chatbot), adversarial search (game-playing AI), and personalization (recommendation systems). The problem, therefore, was to design and implement working solutions for each of these tasks using Python.

4.4 Project Objectives

- To build a rule-based chatbot capable of responding to user queries using pattern matching.
- To implement an unbeatable Tic-Tac-Toe AI agent using the Minimax algorithm with Alpha-Beta Pruning.

- To design a simple recommendation system that suggests movies/books/products based on user preferences.
- To gain practical understanding of Natural Language Processing, Game Theory and Search Algorithms, and Recommendation Techniques (Collaborative and Content-Based Filtering).

4.5 Scope of the Project

The scope of this internship was limited to building foundational, console-based AI applications in Python that demonstrate core concepts rather than production-grade, deployable systems. The chatbot handles a predefined set of rules/patterns; the Tic-Tac-Toe AI is scoped to the standard 3x3 board game; and the recommendation system works on a small sample dataset of user preferences. These projects can further be extended with larger datasets, machine learning models, and web-based interfaces in future work.

4.6 Technologies and Tools Used

- Programming Language: Python 3
- Concepts: If-else / Pattern Matching, Minimax Algorithm, Alpha-Beta Pruning, Collaborative & Content-Based Filtering
- Libraries (as applicable): re (regular expressions), pandas/numpy for data handling in the recommendation system
- Tools: VS Code / Jupyter Notebook, Git & GitHub for version control

4.7 System Architecture

Each of the three tasks follows a simple, modular architecture consisting of an input layer (user query/board state/user preference), a processing/logic layer (rule engine, Minimax search tree, or similarity/filtering engine), and an output layer that returns the chatbot's response, the AI's next move, or the list of recommended items respectively. Since these are independent console-based applications, there is no shared architecture between them beyond the common Python environment used for development.

4.8 Methodology

Task 1: Chatbot with Rule-Based Responses

A simple chatbot was built that responds to user inputs based on predefined rules. If-else statements and pattern matching techniques were used to identify user queries and map them to appropriate predefined responses. This task provided a basic understanding of Natural Language Processing concepts and conversation flow design.

Task 2: Tic-Tac-Toe AI

An AI agent was implemented to play the classic game of Tic-Tac-Toe against a human player. The Minimax algorithm, along with Alpha-Beta Pruning for optimization, was used to make the AI player unbeatable. The board was represented as a matrix, and the algorithm recursively evaluated all possible future game states to choose the optimal move at every turn. This project strengthened understanding of game theory and basic adversarial search algorithms.

Task 4: Recommendation System

A simple recommendation system was created that suggests items (movies, books, or products) to users based on their preferences. Both collaborative filtering (recommending items liked by similar users) and content-based filtering (recommending items similar in attributes to what the user already likes) approaches were explored, using a small sample dataset to demonstrate the working of these techniques.

4.9 Expected Outcomes

- A working rule-based chatbot that can hold a basic conversation with the user.
- An unbeatable Tic-Tac-Toe AI agent that never loses against a human opponent.
- A functional recommendation system that suggests relevant items based on user preference data.
- Strengthened practical knowledge of core AI concepts: NLP basics, game theory/search algorithms, and recommendation techniques.

4.10 Certificates of Completion

[Attach the CodSoft internship completion certificate and any related communication/offer emails as proof here.]

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